

SEMESTER-VI

UCS713: MACHINE LEARNING AND IMAGE PROCESSING				
	L	T	P	Cr
	3	0	2	4.0
Course Objective: To learn the Machine Learning techniques in image processing. It offers some of the most cost-effective approaches to automated knowledge acquisition in emerging data-rich disciplines and focuses on the theoretical understanding of these methods, as well as their computational implications.				
Syllabus Introduction: Introduction to Machine Learning and its type: Supervised learning, Unsupervised learning and Reinforcement learning. Introduction to Digital Image Processing: Digital Image Fundamentals, Simple image formation model, image sampling and quantization, basic relationships between pixels, Examples of fields that use digital image processing, fundamental steps in digital image processing, components of image processing system. Introduction to Image Processing: Image Segmentation, Detection of discontinuous, edge linking and boundary detection, thresholding, region-based segmentation, object recognition, Patterns and patterns classes, recognition based on decision-theoretic methods, matching, optimum statistical classifiers, neural networks, structural methods –matching shape numbers, string matching. Data Pre-processing: Need of Data Pre-processing, Data Pre-processing Methods: Data Cleaning, Data Integration, Data Transformation, Data Reduction; Feature Scaling (Normalization and Standardization), Splitting dataset into Training and Testing set. Regression: Need and Applications of Regression, Simple Linear Regression, Multiple Linear Regression and Polynomial Regression, Evaluating Regression Models Performance (RMSE, Mean Absolute Error, Correlation, R Square, Accuracy with acceptable error, scatter plot, etc). Classification: Need and Applications of Classification, Logistic Regression, Decision tree, Tree induction algorithm – split algorithm based on information theory, split algorithm based on Gini index; Random forest classification, Naïve Bayes algorithm; K-Nearest Neighbours (K-NN), Support Vector Machine (SVM), Evaluating Classification Models Performance (Sensitivity, Specificity, Precision, Recall, etc). Clustering: Need and Applications of Clustering, Partitioned methods, Hierarchical methods, Density-based methods. ANN and CNN: Introduction to ANN and CNN, application in image processing and industrial processes. Application of Machine Learning (Mini Projects): Object Detection, Price prediction, Fault prediction, Quality prediction, defect detection, visual inspection, etc.				
Laboratory Work Demonstrate the use of Image Processing Toolbox on MATLAB and demonstrate the use of OpenCV in Python to create interactive image processing applications like image enhancement, image compression, image segmentation, feature extraction etc. Object Detection, Face extraction, Image Colorization, Laboratory Work using Python: Implement data preprocessing, Simple Linear Regression, Multiple Linear Regression, Decision Tree, Random forest classification, Naïve				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

Bayes algorithm; K-Nearest Neighbors (K-NN), Support Vector Machine, k-Means, Apriori algorithm, ANN.

Course Learning Outcomes (CLOs)

The students will be able to:

1. analyze methods and theories in the field of machine learning and provide an introduction to the basic principles, techniques, and applications of machine learning, supervised, unsupervised and reinforcement learning.
2. apply different classification, regression and clustering techniques for numerical data.
3. comprehend different machine learning techniques in image processing.
4. understand the concept of Neural Networks and its implementation in context of Machine Learning.

Text Books

1. Mitchell M., T., Machine Learning, McGraw Hill (1997) 1st ed.
2. Alpaydin E., Introduction to Machine Learning, MIT Press (2014) 3rd ed.
3. Vijay vargia Abhishek, Machine Learning with Python, BPB Publication (2018).
4. Gonzalez C. R., Woods E. R., Digital Image Processing, Pearson Education (2008) 3rd ed.
5. Sonka M., Hlavac V. and Boyle R., Image Processing, Analysis and Machine Vision, Thomson Learning, (1993) 1st ed.

Reference Books

1. Bishop M., C., Pattern Recognition and Machine Learning, Springer-Verlag (2011) 2nd ed.
2. Michie D., Spiegelhalter J. D., Taylor C. C., Campbell, J., Machine Learning, Neural and Statistical Classification. Overseas Press (1994).
3. McAndrew A., Introduction to Digital Image Processing with Matlab, Thomson Course Technology (2004)
4. Low A., Introductory Computer Vision and Image Processing, McGraw-Hill (1991), 1st ed.

Evaluation scheme

S N	Evaluation Elements	Weights (%)
1.	MST	25
2.	EST	40
3.	Sessionals (May include Assignments/Projects/ Tutorials/Quiz/Lab evaluations)	35

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